

Implementation of a Basic Blockchain with PoW and Merkle Patricia Tries

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1. Introduction

Blockchain research is continually driving new functionalities and optimizations. As a result, modern blockchains look vastly different from the beginning.

This project is a hands-on implementation of a blockchain designed to isolate and study the core focusing on:

- **Consensus mechanism** (PoW)
- **Merkle Patricia Trie** (MPT)

2. Capabilities

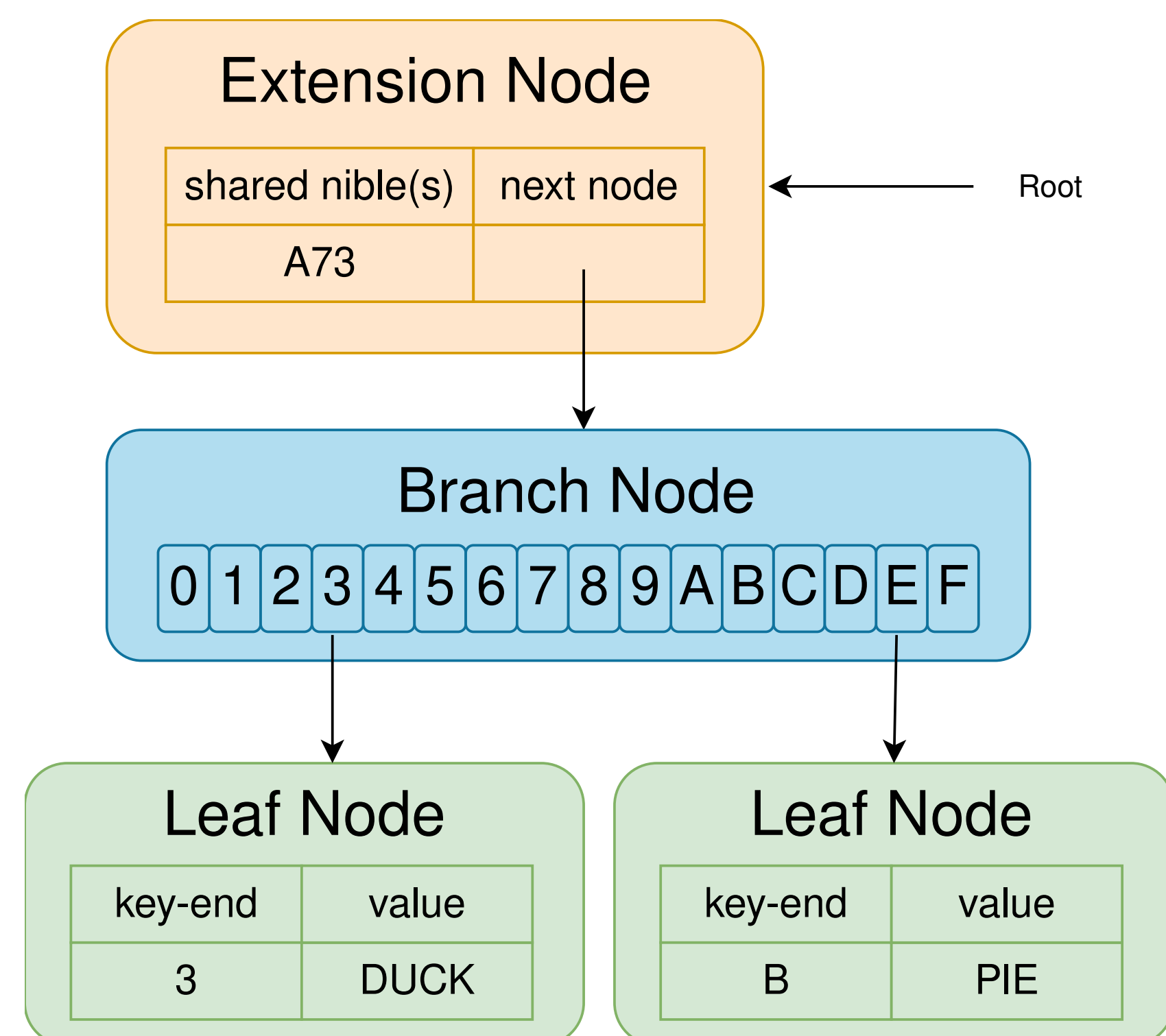
- **Cryptographic Immutability:** Ensuring integrity via sequential SHA-256 block-hashing and Proof-of-Work consensus.
- **Lightweight Verification:** Utilizing Merkle Patricia Trie (MPT) cryptographic proofs $O(\log n)$.
- **Persistent State Storage:** Implementing data-store layer to preserve the blockchain across experimental runs.
- **Debugging tools and tests:** Integrated tools for analyzing the content of blocks in the blockchain and growth.

3. Merkle Patricia Trie (MPT)

Key	Value
0xA7333	DUCK
0xA73EB	PIE

Differences from standard MPT:

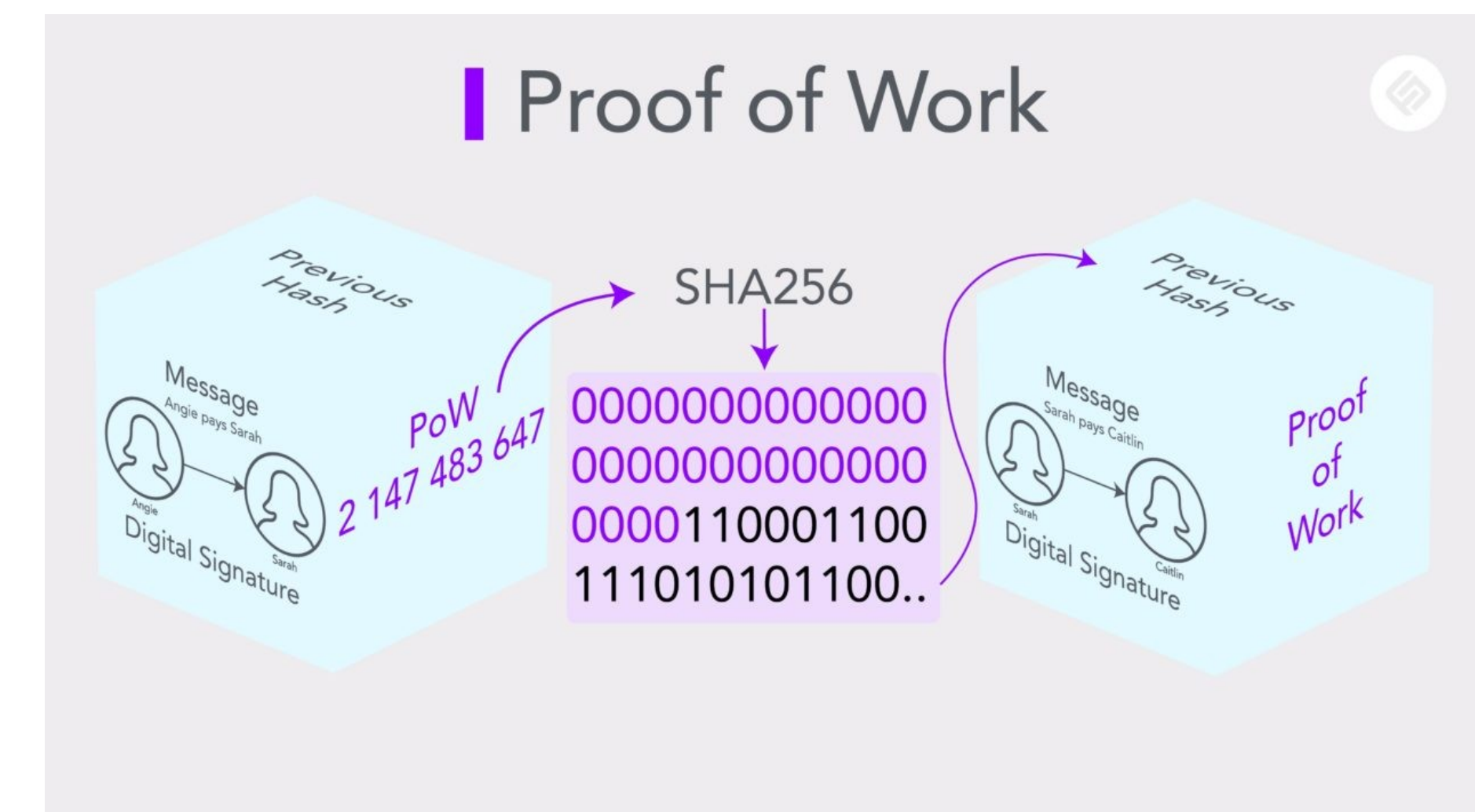
- Use of GOB encoder instead of RLP.
- Branch node does not contain value because all keys have equal length.
- The key of a value (path), is the hash of value itself.
- The values stored are strings transformed to bytes



4. Consensus (PoW)

$$\text{SHA256}(\text{BlockData} \parallel \text{Nonce}) < \text{Target}$$

$$\text{Target} = 1 \ll (\text{HashLength} - \text{Difficulty})$$



5. Class diagram

Architectural design prioritizes modularity and enforces an acyclic dependency graph among classes and defines responsibilities for each component.

